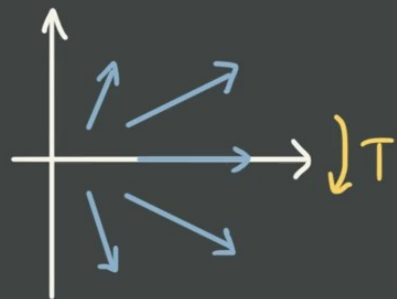


Eigenvalues, eigenvectors, eigenspaces

$$T(\vec{v}) = A\vec{v}$$

$$T(\vec{v}) = \lambda \vec{v}$$



$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$\begin{vmatrix} 2 & -4 \\ -1 & -1 \end{vmatrix} = -2 - 4 = -6$$

$$A\vec{v} = \lambda \vec{v}$$

$$\vec{0} = \lambda \vec{v} - A\vec{v}$$

$$\vec{0} = \lambda I_n \vec{v} - A\vec{v}$$

$$\vec{0} = \underbrace{(\lambda I_n - A)}_B \vec{v}$$

$$B\vec{v} = \vec{0}$$

$$|B| = 0$$

$$|\lambda I_n - A| = 0$$

$$\lambda - 3 = 0 \quad \lambda + 2 = 0 \quad \text{C.P.}$$

$$\boxed{\lambda = 3}$$

$$\boxed{\lambda = -2}$$

$$\text{Trace: } 1 + 3 - 2 = 1$$

$$|A| = -6$$

$$T(\vec{v}) = \begin{bmatrix} 2 & -4 \\ -1 & -1 \end{bmatrix} \vec{v}$$

$$\left| \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & -4 \\ -1 & -1 \end{bmatrix} \right|$$

$$\left| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 2 & -4 \\ -1 & -1 \end{bmatrix} \right|$$

$$\rightarrow \begin{vmatrix} \lambda - 2 & 4 \\ 1 & \lambda + 1 \end{vmatrix}$$

$$(\lambda - 2)(\lambda + 1) - (4)(1)$$

$$\lambda^2 + \lambda - 2\lambda - 2 - 4$$

$$\lambda^2 - \lambda - 6$$

$$\lambda^2 - \lambda - 6 = 0 \quad \text{C.E.}$$

$$(\lambda - 3)(\lambda + 2) = 0$$

$$E_\lambda = N(\lambda I_n - A)$$

$$E_{-2} = N\left(\begin{bmatrix} -4 & 4 \\ 1 & -1 \end{bmatrix}\right)$$

$$\begin{bmatrix} 1 & -1 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix} \quad \begin{matrix} v_1 - v_2 = 0 \\ v_1 = v_2 \\ v_2 = t \end{matrix}$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

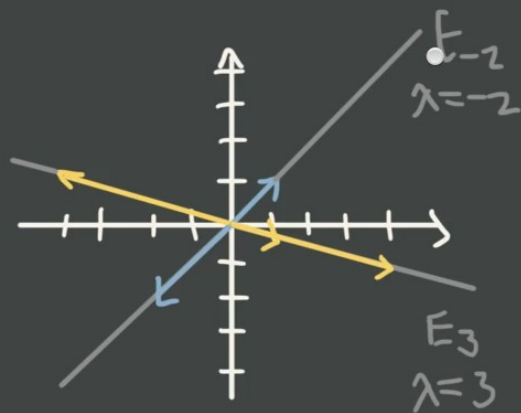
$$E_{-2} = \text{Span}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$$

$$E_3 = N\left(\begin{bmatrix} 1 & 4 \\ 1 & 4 \end{bmatrix}\right)$$

$$\begin{bmatrix} 1 & 4 & | & 0 \\ 0 & 0 & | & 0 \end{bmatrix} \quad \begin{matrix} v_1 + 4v_2 = 0 \\ v_1 = -4v_2 \\ v_2 = t \end{matrix}$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \cdot$$

Eigenvalues, eigenvectors, eigenspaces



$$A\vec{v} = \lambda\vec{v}$$

$$\vec{0} = \lambda\vec{v} - A\vec{v}$$

$$\vec{0} = \lambda I_n \vec{v} - A\vec{v}$$

$$\vec{0} = \underbrace{(\lambda I_n - A)}_B \vec{v}$$

$$B\vec{v} = \vec{0}$$

$$|B| = 0$$

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$$\lambda - 3 = 0 \quad \lambda + 2 = 0 \quad \text{C.P.}$$

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$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$\begin{vmatrix} 2 & -4 \\ -1 & -1 \end{vmatrix} = -2 - 4 = -6$$

$$T(\vec{v}) = \begin{bmatrix} 2 & -4 \\ -1 & -1 \end{bmatrix} \vec{v}$$

$$\left| \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & -4 \\ -1 & -1 \end{bmatrix} \right|$$

$$\left| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} 2 & -4 \\ -1 & -1 \end{bmatrix} \right|$$

$$\rightarrow \begin{vmatrix} \lambda - 2 & 4 \\ 1 & \lambda + 1 \end{vmatrix}$$

$$(\lambda - 2)(\lambda + 1) - (4)(1)$$

$$\lambda^2 + \lambda - 2\lambda - 2 - 4$$

$$\lambda^2 - \lambda - 6$$

$$\lambda^2 - \lambda - 6 = 0 \quad \text{C.E.}$$

$$(\lambda - 3)(\lambda + 2) = 0$$

$$E_\lambda = N(\lambda I_n - A)$$

$$E_{-2} = N\left(\begin{bmatrix} -4 & 4 \\ 1 & -1 \end{bmatrix}\right)$$

$$\begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} v_1 - v_2 = 0 \\ v_1 = v_2 \\ v_2 = t \end{matrix}$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$E_{-2} = \text{span}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right)$$

$$E_3 = N\left(\begin{bmatrix} 1 & 4 \\ 1 & 4 \end{bmatrix}\right)$$

$$\begin{bmatrix} 1 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} v_1 + 4v_2 = 0 \\ v_1 = -4v_2 \\ v_2 = t \end{matrix}$$

$$\begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = t \begin{bmatrix} -4 \\ 1 \end{bmatrix}$$

$$E_3 = \text{span}\left(\begin{bmatrix} -4 \\ 1 \end{bmatrix}\right)$$

Eigen in three dimensions

$$T(\vec{v}) = \begin{bmatrix} 4 & 0 & 1 \\ -1 & -b & -2 \\ 5 & 0 & 0 \end{bmatrix} \vec{v}$$

$$\lambda = -b, -1, 5$$

$$E_{-b} = \left[\begin{array}{ccc|c} -10 & 0 & -1 & 0 \\ 1 & 0 & 2 & 0 \\ -5 & 0 & -b & 0 \end{array} \right]$$

$$|\lambda I_n - A| = \left| \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 4 & 0 & 1 \\ -1 & -b & -2 \\ 5 & 0 & 0 \end{bmatrix} \right|$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ -10 & 0 & -1 & 0 \\ -5 & 0 & -b & 0 \end{array} \right]$$

$$\begin{matrix} v_1 & v_2 & v_3 \\ \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{matrix}$$

$$v_1 = 0 \\ v_3 = 0$$

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$= \left| \begin{bmatrix} \lambda-4 & 0 & -1 \\ 1 & \lambda+b & 2 \\ -5 & 0 & \lambda \end{bmatrix} \right|$$

$$= -0 \left| \begin{array}{c} \diagup \end{array} \right| + (\lambda+b) \left| \begin{array}{cc} \lambda-4 & -1 \\ -5 & \lambda \end{array} \right| - 0 \left| \begin{array}{c} \diagup \end{array} \right|$$

$$= (\lambda+b) [(\lambda-4)\lambda - (-1)(-5)]$$

$$= (\lambda+b) [\lambda^2 - 4\lambda - 5] = (\lambda+b)(\lambda-5)(\lambda+1) = 0$$

Eigen in three dimensions

$$T(\vec{v}) = \begin{bmatrix} 4 & 0 & 1 \\ -1 & -6 & -2 \\ 5 & 0 & 0 \end{bmatrix} \vec{v}$$

$$\lambda = -6, -1, 5$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} -1/5 \\ -9/25 \\ 1 \end{bmatrix}$$

$$E_5 = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 11 & 2 \\ -5 & 0 & 5 \end{bmatrix}$$

$$|\lambda I_n - A| = \left| \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 4 & 0 & 1 \\ -1 & -6 & -2 \\ 5 & 0 & 0 \end{bmatrix} \right|$$

$$= \left| \begin{bmatrix} \lambda-4 & 0 & -1 \\ 1 & \lambda+6 & 2 \\ -5 & 0 & \lambda \end{bmatrix} \right|$$

$$2 - 5 \left(\frac{9}{25} \right) 5$$

$$2 - \frac{9}{5}$$

$$\frac{10}{5} - \frac{9}{5} = \frac{1}{5}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1/5 & 0 \\ 0 & 1 & 9/25 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$v_1 + \frac{1}{5}v_3 = 0$$

$$v_2 + \frac{9}{25}v_3 = 0$$

$$v_1 = -\frac{1}{5}v_3$$

$$v_2 = -\frac{9}{25}v_3$$

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_3 \begin{bmatrix} -1/5 \\ -9/25 \\ 1 \end{bmatrix}$$

Eigen in three dimensions

$$T(\vec{v}) = \begin{bmatrix} 4 & 0 & 1 \\ -1 & -6 & -2 \\ 5 & 0 & 0 \end{bmatrix} \vec{v}$$

$$\lambda = -6, -1, 5$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} -1/5 \\ -9/25 \\ 1 \end{bmatrix} \begin{bmatrix} ! \\ \bullet \\ \bullet \end{bmatrix}$$

$$|\lambda I_n - A| = \left| \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 4 & 0 & 1 \\ -1 & -6 & -2 \\ 5 & 0 & 0 \end{bmatrix} \right|$$

$$= \left| \begin{bmatrix} \lambda-4 & 0 & -1 \\ 1 & \lambda+6 & 2 \\ -5 & 0 & \lambda \end{bmatrix} \right|$$

$$2 - 5 \left(\frac{9}{25} \right) 5$$

$$2 - \frac{9}{5}$$

$$\frac{10}{5} - \frac{9}{5} = \frac{1}{5}$$

$$\begin{bmatrix} 1 & 0 & 1/5 & 0 \\ 0 & 1 & 9/25 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$v_1 + \frac{1}{5} v_3 = 0$$

$$v_2 + \frac{9}{25} v_3 = 0$$

$$v_1 = -\frac{1}{5} v_3$$

$$v_2 = -\frac{9}{25} v_3$$

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_3 \begin{bmatrix} -1/5 \\ -9/25 \\ 1 \end{bmatrix}$$

$$E_5 = \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 1 & 11 & 2 & 0 \\ -5 & 0 & 5 & 0 \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 11 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 3/11 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$v_1 - v_3 = 0$$

$$v_2 + \frac{3}{11} v_3 = 0$$

$$v_1 = v_3$$

$$v_2 = -\frac{3}{11} v_3$$

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_3 \begin{bmatrix} 1 \\ -3/11 \\ 1 \end{bmatrix}$$

Eigen in three dimensions

$$T(\vec{v}) = \begin{bmatrix} 4 & 0 & 1 \\ -1 & -6 & -2 \\ 5 & 0 & 0 \end{bmatrix} \vec{v}$$

$$\lambda = -6, -1, 5$$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} -1/5 \\ -9/25 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ -3/11 \\ 1 \end{bmatrix}$$

$$E_{-6} = \text{span} \left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right)$$

$$E_{-1} = \text{span} \left(\begin{bmatrix} -1/5 \\ -9/25 \\ 1 \end{bmatrix} \right)$$

$$E_5 = \text{span} \left(\begin{bmatrix} 1 \\ -3/11 \\ 1 \end{bmatrix} \right)$$

$$|\lambda I_n - A| = \left| \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} - \begin{bmatrix} 4 & 0 & 1 \\ -1 & -6 & -2 \\ 5 & 0 & 0 \end{bmatrix} \right|$$

$$= \left| \begin{bmatrix} \lambda-4 & 0 & -1 \\ 1 & \lambda+6 & 2 \\ -5 & 0 & \lambda \end{bmatrix} \right|$$

$$2 - 5 \left(\frac{9}{25} \right) 5$$

$$2 - \frac{9}{5}$$

$$\frac{10}{5} - \frac{9}{5} = \frac{1}{5}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1/5 & 0 \\ 0 & 1 & 9/25 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$v_1 + \frac{1}{5} v_3 = 0$$

$$v_2 + \frac{9}{25} v_3 = 0$$

$$v_1 = -\frac{1}{5} v_3$$

$$v_2 = -\frac{9}{25} v_3$$

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_3 \begin{bmatrix} -1/5 \\ -9/25 \\ 1 \end{bmatrix}$$